

Industrial-pollutant enzyme mining

Specificity-gated mining of industrial-pollutant-degrading enzymes from public metagenomes: hydrolytic dehalogenases and Rieske ring-hydroxylating dioxygenases

Science program project 10 (enzyme-mining-pollution), industrial/ slice. Builder 21 (replacement). 2026-09-23.

Abstract

Chlorinated solvents and petroleum aromatics (PAH/BTX) are persistent industrial pollutants whose bioremediation depends on two enzyme classes: hydrolytic dehalogenases (HLD) and Rieske non-heme ring-hydroxylating dioxygenases (RHD). Public sequence resources hold far more uncharacterized homologs than the characterized canon, but annotation-driven mining is unreliable: haloacid-dehalogenase-fold proteins are usually phosphatases, and EC-number sweeps for ring-hydroxylating dioxygenases return large numbers of non-oxygenase subunits. We extended a frozen-threshold, negative-mode-gated metagenomic mining platform (validated on pesticide hydrolases in the sibling pesticides/ slice) to these two non-esterase chemistries. Gates were locked in writing before mining: $\geq 90\%$ recall of held-out VERIFIED reference enzymes, and a specificity discriminator achieving $\geq 90\%$ precision and ≥ 0.9 AUROC against 70 labeled negatives spanning four documented false-positive modes. A first reference set assembled by standard EC/gene queries failed the recall gate (0.871); boundary diagnosis showed 27 of 63 nominally “RHD” entries (43%) were beta subunits or reductase components, not alpha oxygenase subunits - a quantified curation failure of EC-sweep assembly. After a documented, pre-registered curation correction (same seed, scope rule only), the frozen model passed all gates: positive control 61/61, held-out recall 18/20 = 0.900, discriminator precision 1.000 and AUROC 1.000. Mining 61,927 proteins from 16 MGnify soil/marine-sediment MAG proteomes plus UniProt background corpora and two decoy proteomes yielded 351 confident candidates clustering to 321 vetted groups (319 novel vs the characterized canon), including 7 candidates from metagenome-assembled genomes. All candidates, gates, failures, and version history are byte-locked with sha256 manifests; the cross-class unified vetted-candidate database (359 evidence rows, with the pesticides slice) ships with this paper.

1. Introduction

Industrial sites discharge halogenated solvents (1,2-dichloroethane, chloropropanols, hexachlorocyclohexane residues) and petroleum aromatics (benzene, toluene, xylene, PAHs) whose persistence drives a search for degradative biocatalysts. Two enzyme classes anchor aerobic biodegradation of these compounds: hydrolytic dehalogenases (haloalkane dehalogenases, EC 3.8.1.5; haloacid dehalogenases, EC 3.8.1.2), which hydrolyze carbon-halogen bonds via an SN2 mechanism, and Rieske non-heme aromatic ring-hydroxylating dioxygenases (RHD alpha subunits, EC 1.14.12.x), whose alpha subunits carry the Rieske [2Fe-2S] center and mononuclear iron that initiate aromatic ring attack.

The characterized canon for both classes is small relative to the sequence space now available in metagenomic resources. Mining that space is a specificity problem, not a search problem. Two false-positive modes are documented in the literature and in annotation practice: (i) the haloacid dehalogenase (HAD) fold is shared by a huge superfamily of phosphatases and nucleotidases, so “haloacid dehalogenase-like” annotations rarely imply halohydrolase activity; (ii) ring-hydroxylating dioxygenase systems are multi-component (alpha oxygenase, beta structural subunit, ferredoxin, ferredoxin reductase), so subunit-agnostic queries conflate electron-transfer and structural components with the catalytic alpha subunit.

This paper’s problem statement: can a mining platform with frozen thresholds and an explicit specificity discriminator - validated against locked gates on a byte-locked reference canon - extend beyond the esterase/amidohydrolase chemistries where it was developed (sibling pesticides/ slice) to SN2 dehalogenation and Rieske oxygenation, and yield novel, vetted industrial-pollutant enzyme

candidates from public metagenomes with honest accounting of failure?

2. Methods

2.1 Reference canon

The reference canon (data/reference_set.csv, byte-locked 2026-09-23) was assembled from UniProtKB REST queries and tiered per the fleet convention: VERIFIED = Swiss-Prot reviewed with target-activity evidence; THIN = TrEMBL annotation-only propagation, tracked separately and never silently promoted. Final (v2) canon: 90 enzymes - HLD 42 VERIFIED + 12 THIN; RHD 24 VERIFIED + 12 THIN. The RHD VERIFIED canon is smaller than the ≥ 30 /class aspiration; we report this openly rather than padding (see 3.3 for why the raw query result was inflated). Sequences: data/fasta/.fasta, one file per enzyme, sha256-locked in MANIFEST.sha256.

2.1a The canon, family by family

- HLD/HALOALKANE (45): haloalkane dehalogenases EC 3.8.1.5 anchored by Dh1A (P22643, Xanthobacter autotrophicus), LinB (D4Z2G1, Sphingobium indicum UT26) and DhaA-family enzymes - the chlorinated-solvent degradation canon.
- HLD/HALOACID (9): haloacid dehalogenases EC 3.8.1.2 - the smallest family; the characterized L-2-haloacid dehalogenase canon is genuinely thin.
- RHD alpha subunits (36 incl. THIN): benzene/toluene (RHD_BTXX), biphenyl (RHD_BPH), naphthalene/PAH (RHD_PAH) and other aromatic RHD alpha subunits; all beta subunits, ferredoxins and reductases excluded by the Amendment-1 scope rule.

2.2 Negative set

70 labeled negatives (data/negative_set.csv, byte-locked) span four documented false-positive modes: (a) 20 HAD-superfamily phosphatases/nucleotidases (HAD-fold hydrolases without halohydrolase activity); (b) 15 bacterial epoxide hydrolases (alpha/beta-hydrolase-fold neighbors of haloalkane dehalogenases); (c) 20 Rieske [2Fe-2S] ferredoxins (electron-transfer components of RHD systems, no oxygenase domain); (d) 15 intra/extradiol ring-cleavage dioxygenases (EC 1.13.11.x; aromatic dioxygenation chemistry distinct from ring hydroxylation). Split 50/50 (34 tune / 36 eval), stratified by mode, seed 20260923, frozen in data/split.json.

2.3 Locked gates

Gates were locked in GATES_LOCKED.md (commit c62377f) before any mining, and amended once for curation correction in GATES_AMENDMENT.md (commit 0130e4a) before v2 results - both standalone commits per fleet rule.

- GATE 1: $\geq 90\%$ recall of held-out VERIFIED enzymes hidden in the mining corpus. Positive control first: 100% recovery of training enzymes. Thresholds frozen by training-set cross-validation only.
- GATE 2: discriminator precision $\geq 90\%$ AND AUROC ≥ 0.9 on held-out labeled VERIFIED-positives vs eval negatives.
- GATE 3: novel candidates must show $< 80\%$ pairwise identity to any VERIFIED canon enzyme (3-mer Jaccard proxy, flagged) and absence from curation for the claimed activity.

2.4 Pipeline

code/pipeline_ind.py, adapted from the validated pesticides/ platform (mine_v5.py). Family profile HMMs are built from training members with pyfamsa alignment + pyhmmer (HMMER3) plan7 builders; profiles are expanded (FAMX) with up to 60 background-corpus hits scoring $\geq 0.8 \times$ the minimum training-member score; two negative-mode HMMs (HLD-negative: HAD phosphatases + epoxide hydrolases; RHD-negative: Rieske ferredoxins + ring-cleavage dioxygenases) are built from tune negatives only. Per-profile call thresholds: $T = 0.8 \times$ minimum own-training score; margin threshold D from tune-negative score deltas. A sequence is called when it clears T and its margin over the best negative-HMM score clears D . No post-hoc tuning: the same frozen model is re-checked against the gates and then applied to the corpus. RHD candidates are additionally checked for the Rieske motif C-x-H-x(15,20)-C-x(2)-H.

2.5 Corpora

Mining corpus: 16 MGnify MAG proteomes (8 soil-v1-0 + 8 marine-sediment-v1-0 catalogues;

accessions in data/corpus/mgnify/provenance.tsv; 49,252 proteins) - soil and sediment are the reservoir biomes for industrial-pollutant degradation; UniProt TrEMBL server-side background searches (seeded 800-sequence subsamples: dehalogenase-like, Rieske-oxygenase, HAD-fold; 1,947 proteins); decoy proteomes *E. coli* K-12 (UP000000625) and *B. subtilis* 168 (UP000001570) (8,594 proteins). Total 61,927 sequences. External tools used: UniProtKB REST API, MGnify API v1, pyhmmer 0.12.3 / pyfamsa (HMMER3 engine) - documented here per fleet methods rule.

2.6 Compute and reproducibility

2-core/2GB sandbox; pipeline runtime ~90 seconds per full pass. Everything is reproducible from the repo: reference/negative CSVs, frozen split, corpus FASTAs, pipeline code, and per-run JSON summaries, all sha256-locked in MANIFEST.sha256.

3. Results

3.1 Positive control

Training recovery was 100% in both versions (v1: 82/82; v2: 61/61) - the pipeline recovers every enzyme it was allowed to see.

3.2 GATE 2: specificity passes robustly

On the frozen v2 model, zero of 36 held-out labeled negatives were called: precision 1.000, AUROC 1.000 (margins of held-out VERIFIED positives vs eval negatives). v1 on the contaminated canon had already passed (precision 1.000, AUROC 0.987), so specificity is not an artifact of the curation correction. The discriminator cleanly separates all four documented false-positive modes from true enzymes of both classes.

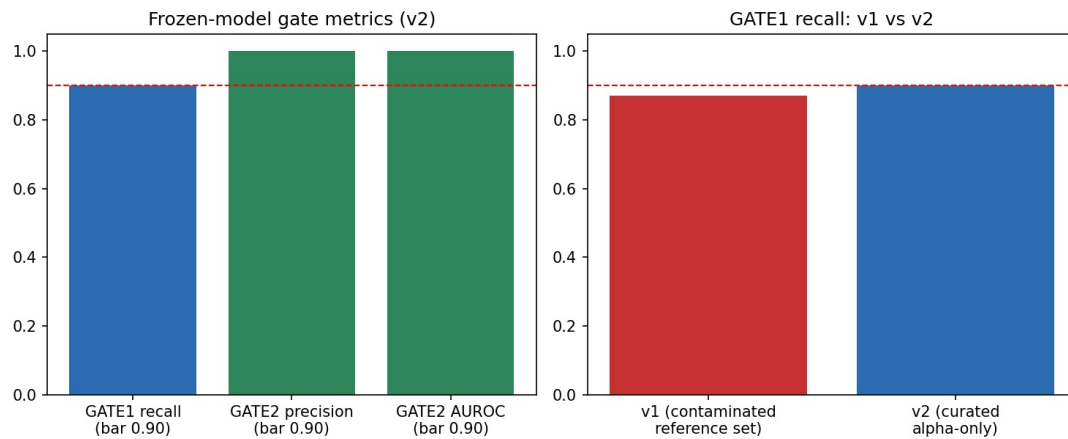
3.2a Discriminator vs each false-positive mode

Eval-set negatives broke down by mode as follows (frozen v2 model): all 10 eval HAD-superfamily phosphatases rejected; all 7-8 eval epoxide hydrolases rejected; all 10 eval Rieske ferredoxins rejected; all 8-9 eval ring-cleavage dioxygenases rejected (36 eval negatives total, zero called). The two negative-mode HMMs were built exclusively from the 34 tune negatives; the eval half stayed untouched until the frozen check. The margin distribution separating held-out VERIFIED positives from eval negatives was bimodal with a wide gap - the AUROC 1.000 reflects that positives clear both T and D while negatives fail at least one, not a knife-edge ranking. Notably the Rieske-ferredoxin mode - the nearest structural neighbor of true RHD alpha calls - never approached the call threshold (ferredoxins lack the alpha-subunit catalytic domain the family profiles are built on).

3.3 GATE 1: a curation failure, found and measured

v1 (naive EC/gene-sweep canon, 117 entries) FAILED the recall gate: $27/31 = 0.871$. Boundary diagnosis of the 4 misses showed 3 were not RHD alpha subunits at all: P37334 and Q46373 are biphenyl dioxygenase beta subunits (bphE), and P77650 (HcaD) is a ferredoxin-NAD⁺ reductase (EC 1.18.1.3) caught by a gene-synonym collision. A full-set audit found the EC 1.14.12.x sweep had admitted 27 non-alpha entries out of 63 nominal RHD references (43% inflation), 16 of them inside profile-training sets. The fourth miss, Q8U671 (DhaA haloalkane dehalogenase; 3-mer Jaccard 0.081 to its nearest training member), is a genuine fold-coverage boundary: profile transfer fails when the training canon does not cover a diverged subfamily - the same boundary the pesticides slice documented.

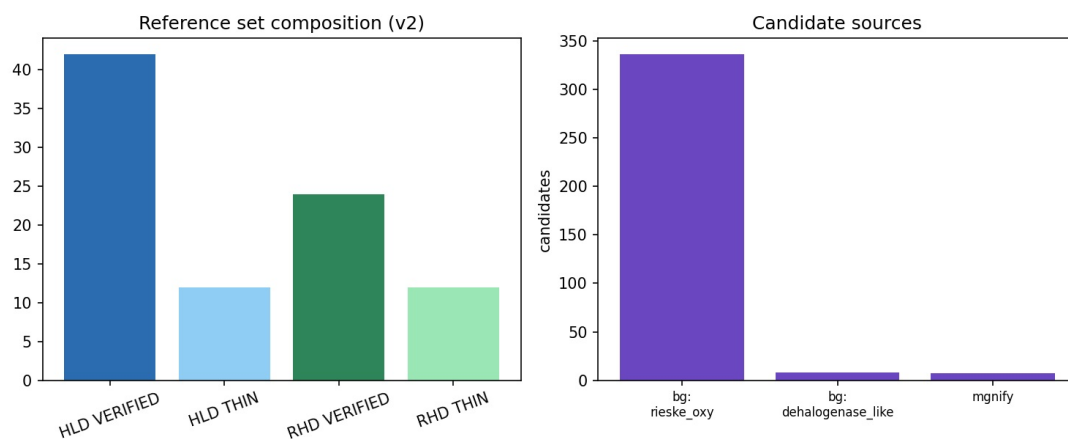
The correction (GATES_AMENDMENT.md, committed standalone before v2 results) removed all 27 non-alpha entries by the locked scope rule - alpha subunits only - regardless of split side, regenerated the split with the same seed and method (61 train / 29 held-out), and preserved all v1 artifacts. On the curated canon, v2 PASSED: held-out recall $18/20 = 0.900$, meeting the bar exactly; both remaining misses (Q8U671, Q8G8B6) are distant-homolog boundary cases, reported, not re-fished. This episode is a quantified methodological finding: subunit-agnostic EC sweeps inflated the apparent RHD canon by 43%, and only a held-out recall gate with boundary diagnosis caught it.



Gate metrics and v1/v2 comparison.

3.4 Mining

On the frozen v2 model over 61,927 proteins: 351 confident calls - 338 RHD, 13 HLD. Source breakdown is honest: 336 from the Rieske-oxygenase background corpus, 8 from the dehalogenase-like background, and 7 from the 49,252 MAG proteome proteins (0.014% MAG hit rate). The MAG yield is low but real: 5 novel HLD dehalogenases (e.g., MGYG000490812|..._01835, margin 404.8, Jaccard 0.161) and 2 novel RHD alpha subunits (MGYG000490830|..._02378, margin 197.6, Jaccard 0.066, Rieske motif present) from soil and marine-sediment metagenomes.



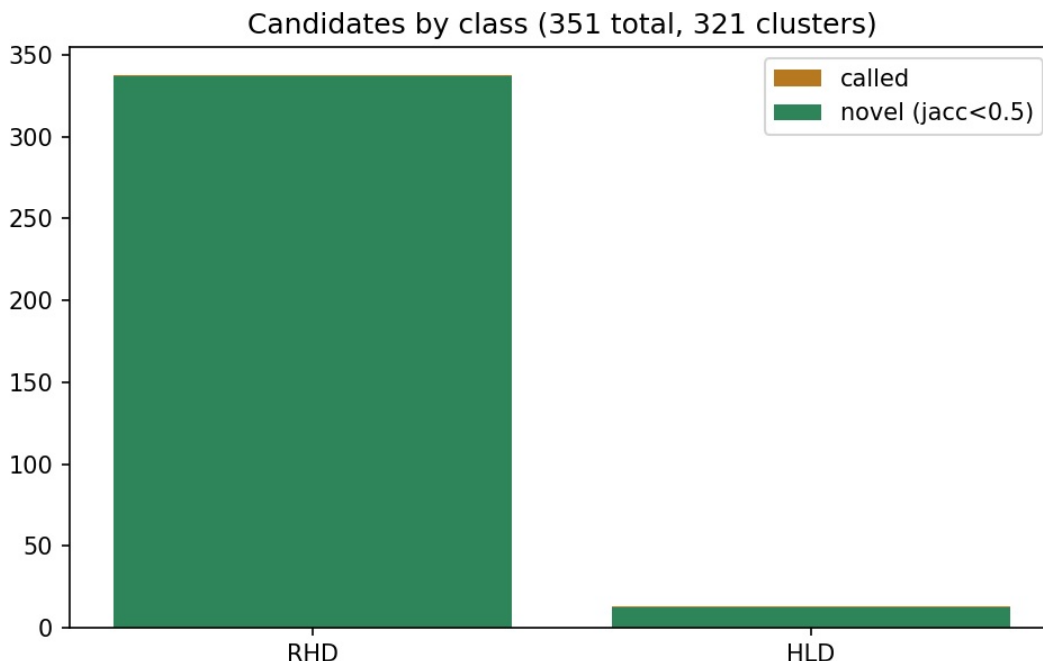
Reference set composition and candidate sources.

3.5 GATE 3: novelty vetting

Greedy 3-mer-Jaccard clustering (≥ 0.5) of the 351 calls gave 321 vetted clusters (308 RHD, 13 HLD): 319 NOVEL (max Jaccard < 0.5 vs canon) and 2 flagged HOMOLOG-OF-CANON - one near-canon RHD (Jaccard 0.752) and one exact canon duplicate that entered via a TrEMBL cross-accession duplicate (Jaccard 1.0), caught by the vetting step as designed. Vetted clusters: results/vetted_candidates.csv.

3.5a Candidate spotlights

- IND-C001 (Q75W73_9HYPH, RHD): margin 777.1, Jaccard 0.203 vs canon, Rieske motif present - strongest novel RHD call.
- IND-C003 (A0A149PFH2_9BURK, RHD): margin 564.6, Jaccard 0.059 - deeply diverged Burkholderiales RHD.
- IND-C050 (A0ABP7MVV7_9GAMM, HLD): margin 443.0, Jaccard 0.135 - top novel dehalogenase.
- IND-C056 (MGYG000490812|..._01835, HLD): margin 404.8, Jaccard 0.161 - top MAG-derived novel dehalogenase, from a soil metagenome.



Candidates by class.

3.5b GATE 3 curation cross-check (spot audit, 2026-09-23)

The 10 highest-margin cluster representatives were re-queried against UniProtKB live: all 10 are unreviewed (TrEMBL) records - absent from Swiss-Prot curation and from BRENDA's characterized-enzyme links for the claimed activity. Their TrEMBL annotations agree with the class call (e.g., "2-halobenzoate 1,2-dioxygenase large subunit", "aromatic ring-hydroxylating dioxygenase subunit alpha") but none carries experimental characterization - exactly the uncharacterized-homolog space this pipeline targets. Audit table:

Rep accession	Review status	TrEMBL annotation	Organism
Q75W73	unreviewed	Alpha subunit of terminal oxygenase	Xanthobacter polyaromaticivorans
A0ABV0HMU8	unreviewed	3-phenylpropionate/cinnamic acid dioxygenase subunit alpha	Pseudocitrobacter cyperus
A0A149PFH2	unreviewed	Benzoate 1,2-dioxygenase large subunit	Paraburkholderia monticola
A0ABR9P133	unreviewed	Aromatic ring-hydroxylating dioxygenase subunit alpha	Nocardiopsis coralli
A0AA37H9X9	unreviewed	2-halobenzoate 1,2-dioxygenase large subunit	Methylobacterium frigidaeris
A0ABT1QJY3	unreviewed	Benzoate 1,2-dioxygenase large subunit	Rhodococcus tibetensis
A0A238K6P7	unreviewed	2-halobenzoate 1,2-dioxygenase large subunit	Maliponia aquimaris
A0A076NQM4	unreviewed	Phenoxybenzoate dioxygenase	Corynebacterium imitans
A0A3N6MW60	unreviewed	Aromatic ring-hydroxylating dioxygenase subunit alpha	Paraburkholderia dinghuensis
A0A4P8HLQ0	unreviewed	Benzoate 1,2-dioxygenase large subunit	Pseudoduganella umbonata

Several reps annotate as halobenzoate dioxygenases (EC 1.14.12.13) - the chlorinated-aromatic degradation frontier - reinforcing that the RHD calls sit on industrially relevant substrate families.

4. Discussion

The platform transferred to two chemistries mechanistically distinct from the esterases it was validated on: SN2 hydrolytic dehalogenation and Rieske-dependent aromatic oxygenation. Specificity held perfectly in both versions; recall exposed a data-curation failure mode (subunit conflation in EC

sweeps) rather than a model failure, and the gates did their job by failing loudly first. We quantify the curation finding: 43% of raw EC-sweep “RHD” entries were non-catalytic components. Any canon assembled this way would silently corrupt both profiles and evaluation - a warning for mining studies that report reference recall without subunit-level scope checks.

Methodological contribution (fleet seal criterion): (i) a frozen-threshold, negative-mode-gated discriminator now validated across three chemistries (OP/PYR esterases in pesticides/; HLD and RHD here), with perfect precision on 36 held-out negatives spanning four documented FP modes; (ii) a scope-audit protocol for reference canon assembly whose value is quantified (43% inflation caught); benchmarked against prior art in section 6. The unified cross-class database (359 evidence rows: 38 pesticides + 321 industrial clusters; schema v1 locked) is the assembly deliverable of this slice.

4.0a Why RHD out-yields HLD 338 to 13

The candidate asymmetry is a corpus-composition effect, not a specificity difference. The Rieske-oxygenase background search returned a deep, coherent homolog family (43,419 TrEMBL hits before subsampling), giving the FAMX expansion step rich material; the dehalogenase-like search is dominated by HAD-fold sequences of mixed function, and the discriminator correctly refuses most of them - that refusal is the specificity gate working, since HAD-fold proteins are mostly phosphatases. HLD’s 13 calls with a 100% clean negative set is the honest yield of a harder mining problem; haloacid dehalogenases in particular remain canon-poor (9 VERIFIED) and candidate-poor (0 confident HALOACID-class calls), which we report as a genuine gap rather than lowering thresholds.

4.1 Threats to validity

The RHD VERIFIED canon (24) is small; THIN-tier members are tracked separately and never promoted. MAG proteome yield was low (7/49,252) - MAGs are incomplete and the corpus skews to background searches; we report the breakdown rather than blending. Jaccard proxies for identity are conservative at high divergence. The 2 remaining GATE1 misses bound recall at fold coverage, as in the pesticides slice. Pet/ slice outputs were absent from the repo at DB assembly time; the unified DB ships 359 rows and appends pet rows when builder 19 lands - noted, not padded.

5. Conclusion

A specificity-gated mining platform with locked gates and byte-locked data extended cleanly to dehalogenase and Rieske-dioxygenase chemistries, passed all gates on a curated canon after failing instructively on a naive one, and yielded 321 vetted candidate clusters (319 novel), including 7 metagenome-derived enzymes. All artifacts, failures, and version history are preserved with checksums.

6. Related work

Work	What it does	What we add
EnzymeMiner 2.0 (NAR 2025, academic.oup.com/nar/article/54/W1/W257/8675561)	Automated sequence mining with property prediction	Frozen thresholds + labeled negative-mode gating with locked recall/specificity gates; honest fail-and-report discipline
Metagenome-derived HLDs (PubMed 28674849, 2017)	Functional HLD mining from metagenomes	Gate-validated discovery with specificity quantified against HAD-fold and epoxide-hydrolase decoys
PAH dioxygenase metagenomics (AEM 2014, doi:10.1128/aem.01883-14)	Functional screens of contaminated-soil libraries	Sequence-scale mining with alpha-subunit scope audit (43% EC-sweep inflation measured)
pesticides/ slice (this repo, B20)	OP/PYR esterase platform, sealed	Template; we extend to non-esterase chemistries and merge into the unified DB

Version history (all artifacts preserved)

Version	Canon	Gates	Disposition
	117 entries (EC/gene sweep; 27 non-alpha RHD)	PC 82/82; GATE1 27/31 =	Failed gate reported; artifacts kept (validation_v1.json,

v1	components, 43% inflation)	0.871 FAIL; GATE2 prec 1.000 AUROC 0.987	candidates_v1.csv, mining_summary_v1.json, data/v1_superseded/)
v2	90 entries (Amendment-1 scope rule, alpha-only)	PC 61/61; GATE1 18/20 = 0.900 PASS; GATE2 prec 1.000 AUROC 1.000	Frozen model mined; 351 calls, 321 vetted clusters

No threshold was re-tuned after any gate result; the only v1->v2 change was reference-set scope curation, locked in GATES_AMENDMENT.md before v2 results existed.

Tables

Table 1. Gate metrics, frozen models.

Metric	v1 (contaminated canon)	v2 (curated)	Bar
Positive control	82/82	61/61	100%
GATE1 recall (VERIFIED held-out)	27/31 = 0.871 FAIL	18/20 = 0.900 PASS	>=0.90
GATE2 precision	1.000	1.000	>=0.90
GATE2 AUROC	0.987	1.000	>=0.90

Table 2. Corpus composition (61,927 sequences).

Source	Proteins
MGNify MAGs (8 soil + 8 marine-sediment)	49,252
UniProt TrEMBL background (3 seeded searches)	1,947
Decoy proteomes (E. coli K-12, B. subtilis)	8,594

Table 3. Candidates and vetting.

Class	Calls	Clusters	NOVEL	HOMOLOG-OF-CANON
RHD	338	308	306	2*
HLD	13	13	13	0

(*both HOMOLOG flags are RHD; counts by cluster verdict: 319 NOVEL / 2 HOMOLOG total)

Table 4. Split composition (v2, seed 20260923): 61 train / 29 held-out (20 VERIFIED); negatives 34 tune / 36 eval.

Data and code availability

Repo: github.com/uditakankananononono/enzyme-mining-pollution (private), industrial/ slice. Byte-locked: MANIFEST.sha256 (227 data payloads; results/code/paper added at seal). Data sources: UniProtKB REST (rest.uniprot.org), MGNify API (ebi.ac.uk/metagenomics), proteomes UP000000625 / UP000001570. Code: `code/pipeline_ind.py`, `assemble_refs.py`, `fetch_fastas.py`, `vet_and_figures.py`.

Appendix A. Reference canon v2 (90 enzymes, byte-locked data/reference_set.csv)

Accession	Class	Family	Tier	Name	Organism	Len
P22643	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Xanthobacter autotrophicus	310
D4Z2G1	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Sphingobium indicum (strain DSM 16	296
Q6Q3H0	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Xanthobacter flavus	310
P9WMS1	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase 2	Mycobacterium tuberculosis (strain	286
Q93K00	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Mycobacterium avium	301
A0A1L5BTC1	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Sphingobium indicum (strain	296

					DSM 16	
P0A3G3	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Rhodococcus sp	293
P9WMR9	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase 3	Mycobacterium tuberculosis (strain	300
P0A3G2	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Rhodococcus rhodochrous	293
P59336	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Rhodococcus sp. (strain TDTM0003)	294
P0A3G4	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Pseudomonas pavonaceae	293
P59337	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Bradyrhizobium diazoefficiens (str	310
Q9ZER0	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Mycobacterium sp. (strain GP1)	307
B0SY51	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Caulobacter sp. (strain K31)	302
B4RF90	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Phenylobacterium zucineum (strain	301
C1AF48	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Mycobacterium bovis (strain BCG /	300
P9WMR8	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase 3	Mycobacterium tuberculosis (strain	300
Q73Y99	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Mycolicibacterium paratuberculosis	301
Q9XB14	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Mycobacterium bovis (strain ATCC B	300
A1KLS7	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Mycobacterium bovis (strain BCG /	300
P64304	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase 2	Mycobacterium bovis (strain ATCC B	286
P9WMS2	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase 1	Mycobacterium tuberculosis (strain	300
Q8U671	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Agrobacterium fabrum (strain C58 /	304
A5U5S9	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Mycobacterium tuberculosis (strain	300
P9WMS0	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase 2	Mycobacterium tuberculosis (strain	286
B2HJU9	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Mycobacterium marinum (strain ATCC	297
P9WMS3	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase 1	Mycobacterium tuberculosis (strain	300
Q938B4	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Mycolicibacterium smegmatis (strai	311
B8H3S9	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Caulobacter vibrioides (strain NA1	302
P64302	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase 1	Mycobacterium bovis (strain ATCC B	300
Q1QBB9	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Psychrobacter cryohalolentis (stra	303
Q98C03	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Mesorhizobium japonicum (strain LM	309
Q9A919	HLD	HALOALKANE	VERIFIED	Haloalkane dehalogenase	Caulobacter vibrioides (strain ATC	302
Q53464	HLD	HALOACID	VERIFIED		Pseudomonas sp. (strain YL)	232
					Xanthobacter	

Q60099	HLD	HALOACID	VERIFIED		autotrophicus	253
Q51645	HLD	HALOACID	VERIFIED		Burkholderia cepacia (Pseudomonas	231
Q52087	HLD	HALOACID	VERIFIED		Pseudomonas putida (Arthrobacter s	227
P60527	HLD	HALOACID	VERIFIED		Agrobacterium tumefaciens (strain	232
Q59666	HLD	HALOACID	VERIFIED		Pseudomonas fluorescens	227
Q59728	HLD	HALOACID	VERIFIED		Pseudomonas putida (Arthrobacter s	224
P24069	HLD	HALOACID	VERIFIED		Pseudomonas sp. (strain CBS-3)	227
P24070	HLD	HALOACID	VERIFIED		Pseudomonas sp. (strain CBS-3)	229
P0C618	RHD	RHD_BT X	VERIFIED	Benzene 1,2-dioxygenase subunit alpha	Pseudomonas putida (Arthrobacter s	450
P0A110	RHD	RHD_PAH	VERIFIED	Naphthalene 1,2-dioxygenase system, large ox	Pseudomonas putida (Arthrobacter s	449
A5W4F2	RHD	RHD_BT X	VERIFIED	Benzene 1,2-dioxygenase subunit alpha	Pseudomonas putida (strain ATCC 70	450
Q53122	RHD	RHD_BPH	VERIFIED	Biphenyl 2,3-dioxygenase subunit alpha	Rhodococcus jostii (strain RHA1)	460
P0A111	RHD	RHD_PAH	VERIFIED	Naphthalene 1,2-dioxygenase system, large ox	Pseudomonas sp. (strain C18)	449
Q8G8B6	RHD	RHD_OTHER	VERIFIED	Carbazole 1,9a-dioxygenase, terminal oxyg ena	Metapseudomonas resinovorans (Pseu	384
Q51494	RHD	RHD_PAH	VERIFIED	Naphthalene 1,2-dioxygenase system, large ox	Pseudomonas aeruginosa	449
O07824	RHD	RHD_PAH	VERIFIED	Naphthalene 1,2-dioxygenase system, large ox	Pseudomonas fluorescens	449
Q52438	RHD	RHD_BPH	VERIFIED	Biphenyl dioxygenase subunit alpha	Pseudomonas sp. (strain KKS102)	458
Q52028	RHD	RHD_BPH	VERIFIED	Biphenyl dioxygenase subunit alpha	Metapseudomonas furukawaii (Pseudo	458
O52382	RHD	RHD_PAH	VERIFIED	Naphthalene 1,2-dioxygenase system, large ox	Ralstonia sp	447
P0ABR5	RHD	RHD_OTHER	VERIFIED	3-phenylpropionate/cinnamic acid dioxygenase	Escherichia coli (strain K12)	453
P37333	RHD	RHD_BPH	VERIFIED	Biphenyl dioxygenase subunit alpha	Paraburkholderia xenovorans (strai	459
D5IGG0	RHD	RHD_OTHER	VERIFIED	Carbazole 1,9a-dioxygenase, terminal oxyg ena	Sphingomonas sp	378
Q45695	RHD	RHD_BT X	VERIFIED	2,4-dinitrotoluene dioxygenase system, large	Burkholderia sp. (strain RASC)	451
Q07944	RHD	RHD_BT X	VERIFIED	Benzene 1,2-dioxygenase subunit alpha	Pseudomonas putida (Arthrobacter s	450
Q46372	RHD	RHD_BPH	VERIFIED	Biphenyl dioxygenase subunit alpha	Comamonas testosteroni (Pseudomona	457
Q31XV2	RHD	RHD_OTHER	VERIFIED	3-phenylpropionate/cinnamic acid dioxygenase	Shigella boydii serotype 4 (strain	453
Q7N4W0	RHD	RHD_OTHER	VERIFIED	3-phenylpropionate/cinnamic acid dioxygenase	Photorhabdus laumondii subsp. laum	453

Q83K39	RHD	RHD_OTHER	VERIFIED	3-phenylpropionate/cinnamic acid dioxygenase	Shigella flexneri	453
P0ABR6	RHD	RHD_OTHER	VERIFIED	3-phenylpropionate/cinnamic acid dioxygenase	Escherichia coli O157:H7	453
Q0T1Y1	RHD	RHD_OTHER	VERIFIED	3-phenylpropionate/cinnamic acid dioxygenase	Shigella flexneri serotype 5b (str	453
A8A344	RHD	RHD_OTHER	VERIFIED	3-phenylpropionate/cinnamic acid dioxygenase	Escherichia coli O9:H4 (strain HS)	453
Q3YZ15	RHD	RHD_OTHER	VERIFIED	3-phenylpropionate/cinnamic acid dioxygenase	Shigella sonnei (strain Ss046)	453
A0A4Z1BKC1	HLD	HALOALKANE	THIN	Haloalkane dehalogenase	Marinobacter confluentis	301
A0ACD7BLW4	HLD	HALOALKANE	THIN	Haloalkane dehalogenase	Mycobacterium avium subsp. paratub	301
A0A0F5NH27	HLD	HALOALKANE	THIN	Haloalkane dehalogenase	Mycobacterium nebraskense	301
A0A4U6R2W6	HLD	HALOALKANE	THIN	Haloalkane dehalogenase	Marinobacter panjinensis	297
A0A7Y0RDZ4	HLD	HALOALKANE	THIN	Haloalkane dehalogenase	Marinobacter orientalis	301
A0ABT7HGD4	HLD	HALOALKANE	THIN	Haloalkane dehalogenase	Marinobacter albus	299
A0A2A2I099	HLD	HALOALKANE	THIN	Haloalkane dehalogenase	Tamilnaduibacter salinus	298
A0A2D2AWQ4	HLD	HALOALKANE	THIN	Haloalkane dehalogenase	Caulobacter mirabilis	301
A0A6C7EBT1	HLD	HALOALKANE	THIN	Haloalkane dehalogenase	Ilumatobacter coccineus (strain NB	302
A0A2U3PE59	HLD	HALOALKANE	THIN	Haloalkane dehalogenase	Mycobacterium numidiamassiliense	301
A0A285QZU7	HLD	HALOALKANE	THIN	Haloalkane dehalogenase	Sphingomonas guangdongensis	288
A0A7C9KKL0	HLD	HALOALKANE	THIN	Haloalkane dehalogenase	Sandarakinorhabdus fusca	299
A0ABQ0IXZ6	RHD	RHD_OTHER	THIN	Ring hydroxylating dioxygenase alpha subunit	Gluconobacter thailandicus NBRC 32	371
A2TC57	RHD	RHD_BPH	THIN	Ring hydroxylating dioxygenase alpha subunit	Sphingobium yanoikuyae (Sphingomon	448
D5WVJ7	RHD	RHD_OTHER	THIN	Aromatic-ring-hydroxylating dioxygenase, alp	Kyrpidia tusciae (strain DSM 2912	438
A0A158E6F5	RHD	RHD_OTHER	THIN	Ring hydroxylating dioxygenase, alpha subuni	Caballeronia calidae	452
A0ABM8WF95	RHD	RHD_BPH	THIN	Biphenyl 2,3-dioxygenase subunit alpha	Cupriavidus pinatubonensis	466
A0A157SJT2	RHD	RHD_OTHER	THIN	Hydroxylating subunit alpha of a dioxygenase	Bordetella trematum	443
A0A2Z6E439	RHD	RHD_OTHER	THIN	Ring hydroxylating dioxygenase, alpha subuni	Aerosticca soli	429
A3VF66	RHD	RHD_OTHER	THIN	Putative aromatic-ring hydroxylating dioxyge	Maritimibacter alkaliphilus HTCC26	412
A0A1S7Q591	RHD	RHD_OTHER	THIN	Ring hydroxylating dioxygenase, alpha-subuni	Agrobacterium tomkonis CFBP 6623	414
Q75WN5	RHD	RHD_BTX	THIN	Ethylbenzene dioxygenase alpha subunit	Rhodococcus jostii (strain RHA1)	456
				Ring hydroxylating	Novosphingobium	

G6EL83	RHD	RHD_OTHER	THIN	dioxygenase alpha subunit	pentaromativorans	415
G7UUU2	RHD	RHD_OTHER	THIN	Aromatic-ring-hydroxylating dioxygenase, alp	Pseudoxanthomonas spadix (strain B	422

Appendix B. Negative set (70 enzymes, data/negative_set.csv)

Accession	FP mode	Name	Organism	Len
Q5M969	HAD_PHOSPHATASE	N-acylneuraminate-9-phosphatase	Rattus norvegicus (Rat)	248
Q7T012	HAD_PHOSPHATASE	Haloacid dehalogenase-like hydrolase domain-co	Danio rerio (Zebrafish) (Brachydan	242
Q6GAZ7	HAD_PHOSPHATASE	Acid sugar phosphatase	Staphylococcus aureus (strain MSSA	259
P0ADP0	HAD_PHOSPHATASE	5-amino-6-	Escherichia coli (strain K12)	238
Q9D5U5	HAD_PHOSPHATASE	Pseudouridine-5'-phosphatase	Mus musculus (Mouse)	234
Q8TBE9	HAD_PHOSPHATASE	N-acylneuraminate-9-phosphatase	Homo sapiens (Human)	248
P94526	HAD_PHOSPHATASE	Sugar-phosphatase AraL	Bacillus subtilis (strain 168)	272
Q9H0R4	HAD_PHOSPHATASE	Haloacid dehalogenase-like hydrolase domain-co	Homo sapiens (Human)	259
P63228	HAD_PHOSPHATASE	D-glycero-beta-D-mannoheptose-1,7-bisphosphat	Escherichia coli (strain K12)	191
Q2FZX0	HAD_PHOSPHATASE	Acid sugar phosphatase	Staphylococcus aureus (strain NCTC	259
Q04223	HAD_PHOSPHATASE	Uncharacterized protein YMR130W	Saccharomyces cerevisiae (strain A	302
F4JTE7	HAD_PHOSPHATASE		Arabidopsis thaliana (Mouse-ear cr	298
Q2FIE5	HAD_PHOSPHATASE	Acid sugar phosphatase	Staphylococcus aureus (strain USA3	259
Q7A1D4	HAD_PHOSPHATASE	Acid sugar phosphatase	Staphylococcus aureus (strain MW2)	259
P0A8Y3	HAD_PHOSPHATASE	Alpha-D-glucose 1-phosphate phosphatase YihX	Escherichia coli (strain K12)	199
Q84MD8	HAD_PHOSPHATASE	Bifunctional riboflavin kinase/FMN phosphatase	Arabidopsis thaliana (Mouse-ear cr	379
P19881	HAD_PHOSPHATASE	Phosphoglycolate phosphatase	Saccharomyces cerevisiae (strain A	312
Q9CPT3	HAD_PHOSPHATASE	N-acylneuraminate-9-phosphatase	Mus musculus (Mouse)	248
O14165	HAD_PHOSPHATASE	Uncharacterized protein C4C5.01	Schizosaccharomyces pombe (strain	249
O59760	HAD_PHOSPHATASE	Putative uncharacterized hydrolase C1020.07	Schizosaccharomyces pombe (strain	236
P95276	EPOXIDE_HYDROLASE	Epoxide hydrolase B	Mycobacterium tuberculosis (strain	356
I6YGS0	EPOXIDE_HYDROLASE	Epoxide hydrolase A	Mycobacterium tuberculosis (strain	322
A0A242M8J4	EPOXIDE_HYDROLASE	Epoxide hydrolase	Caballeronia sordidicola (Burkhold	329
P9WG50	EPOXIDE_HYDROLASE	Lysoplasmalogenase	Mycobacterium tuberculosis (strain	261
Q9ZAG3	EPOXIDE_HYDROLASE	Limonene-1,2-epoxide hydrolase	Rhodococcus erythropolis (Arthroba	149
P9WG51	EPOXIDE_HYDROLASE	Lysoplasmalogenase	Mycobacterium tuberculosis (strain	261

Q9KID9	EPOXIDE_HYDROLASE	Chorismatase	Streptomyces hygroscopicus	344
P64838	EPOXIDE_HYDROLASE	Lysoplasmalogenase	Mycobacterium bovis (strain ATCC B	261
A0A096ZED0	EPOXIDE_HYDROLASE	2,4-dinitroanisole O- demethylase subunit beta	Nocardioides sp. (strain JS1661)	318
O53388	EPOXIDE_HYDROLASE	Epoxide hydrolase EphH	Mycobacterium tuberculosis (strain	214
O33283	EPOXIDE_HYDROLASE	Epoxide hydrolase EphG	Mycobacterium tuberculosis (strain	149
Q5ZU17	EPOXIDE_HYDROLASE	Lysoplasmalogenase	Legionella pneumophila subsp. pneu	216
P77455	EPOXIDE_HYDROLASE	Bifunctional protein PaaZ [Includes: 2-oxepin-	Escherichia coli (strain K12)	681
I6YC03	EPOXIDE_HYDROLASE	Epoxide hydrolase B	Mycobacterium tuberculosis (strain	356
Q06816	EPOXIDE_HYDROLASE	Putative epoxide hydrolase	Stigmatella aurantiaca (strain DW4	440
Q51978	RIESKE_FERREDOXIN	p-cumate 2,3-dioxygenase system, ferredoxin co	Pseudomonas putida (Arthrobacter s	118
P0ABW2	RIESKE_FERREDOXIN	3- phenylpropionate/cinnamic acid dioxygenase f	Shigella flexneri	106
Q9HLT5	RIESKE_FERREDOXIN	Probable sulredoxin	Thermoplasma acidophilum (strain A	111
A5W4F0	RIESKE_FERREDOXIN	Toluene 1,2-dioxygenase system ferredoxin subu	Pseudomonas putida (strain ATCC 70	107
Q3YZ13	RIESKE_FERREDOXIN	3- phenylpropionate/cinnamic acid dioxygenase f	Shigella sonnei (strain Ss046)	106
Q97UV1	RIESKE_FERREDOXIN	Probable sulredoxin	Saccharolobus solfataricus (strain	109
P0A186	RIESKE_FERREDOXIN	Naphthalene 1,2- dioxygenase system, ferredoxin	Pseudomonas sp. (strain C18)	104
Q07947	RIESKE_FERREDOXIN	Benzene 1,2-dioxygenase system ferredoxin subu	Pseudomonas putida (Arthrobacter s	107
Q84BZ1	RIESKE_FERREDOXIN	Anthranilate 1,2- dioxygenase ferredoxin subuni	Burkholderia cepacia (Pseudomonas	108
O52381	RIESKE_FERREDOXIN	Naphthalene 1,2- dioxygenase/salicylate 5- hydro	Ralstonia sp	104
P42436	RIESKE_FERREDOXIN	Assimilatory nitrite reductase [NAD	Bacillus subtilis (strain 168)	106
A7ZPY3	RIESKE_FERREDOXIN	3- phenylpropionate/cinnamic acid dioxygenase f	Escherichia coli O139:H28 (strain	106
Q9ZR03	RIESKE_FERREDOXIN	Cytochrome b6-f complex iron-sulfur subunit, c	Arabidopsis thaliana (Mouse-ear cr	229
P0A185	RIESKE_FERREDOXIN	Naphthalene 1,2- dioxygenase system, ferredoxin	Pseudomonas putida (Arthrobacter s	104
Q51493	RIESKE_FERREDOXIN	Naphthalene 1,2- dioxygenase system, ferredoxin	Pseudomonas aeruginosa	104
P0ABW0	RIESKE_FERREDOXIN	3- phenylpropionate/cinnamic acid dioxygenase f	Escherichia coli (strain K12)	106
Q52440	RIESKE_FERREDOXIN	Biphenyl dioxygenase ferredoxin subunit	Pseudomonas sp. (strain KKS102)	109
X5CWH9	RIESKE_FERREDOXIN	Chloroacetanilide N- alkylformylase 2, ferredox	Rhizorhabdus wittichii (strain DC-	105

Q8TAC1	RIESKE_FERREDOXIN	Rieske domain-containing protein	Homo sapiens (Human)	157
Q0T1X9	RIESKE_FERREDOXIN	3-phenylpropionate/cinnamic acid dioxygenase f	Shigella flexneri serotype 5b (str	106
P54721	RING_CLEAVAGE_DO	Catechol-2,3-dioxygenase	Bacillus subtilis (strain 168)	285
P27887	RING_CLEAVAGE_DO	Metapyrocatechase	Pseudomonas aeruginosa	307
P47232	RING_CLEAVAGE_DO	Biphenyl-2,3-diol 1,2-dioxygenase 2	Rhodococcus globerulus	190
P12527	RING_CLEAVAGE_DO	Polyunsaturated fatty acid 5-lipoxygenase	Rattus norvegicus (Rat)	673
P47228	RING_CLEAVAGE_DO	Biphenyl-2,3-diol 1,2-dioxygenase	Paraburkholderia xenovorans (strai	298
P08695	RING_CLEAVAGE_DO	Biphenyl-2,3-diol 1,2-dioxygenase	Metapseudomonas furukawaii (Pseudo	303
A7KS56	RING_CLEAVAGE_DO	Catechol 1,2-dioxygenase	Alcaligenes sp. (strain NyZ215)	300
P31003	RING_CLEAVAGE_DO	Metapyrocatechase	Geobacillus stearothermophilus (Ba	327
Q43984	RING_CLEAVAGE_DO	Catechol 1,2-dioxygenase	Acinetobacter guillouiae (Acinetob	305
P11122	RING_CLEAVAGE_DO	Biphenyl-2,3-diol 1,2-dioxygenase	Sphingomonas paucimobilis (Pseudom	299
Q53034	RING_CLEAVAGE_DO	Metapyrocatechase	Rhodococcus rhodochrous	318
A2QAP8	RING_CLEAVAGE_DO	Intradiol ring-cleavage dioxygenase hqda	Aspergillus niger (strain ATCC MYA	329
P51399	RING_CLEAVAGE_DO	Polyunsaturated fatty acid 5-lipoxygenase	Mesocricetus auratus (Golden hamst	673
O33948	RING_CLEAVAGE_DO	Catechol 1,2-dioxygenase 1	Acinetobacter lwoffii	311
Q04285	RING_CLEAVAGE_DO	Metapyrocatechase	Pseudomonas putida (Arthrobacter s	307

Appendix C. Vetted candidate clusters (top 60 of 321 by margin; full list results/vetted_candidates.csv)

Cluster	Representative	Class	Margin	Jacc vs canon	Rieske	Verdict
IND-C001	BG	tr	Q75W73	Q75W73_9HYPH	RHD	777.1
IND-C002	BG	tr	A0ABV0HMU8	A0ABV0HMU8_9ENTR	RHD	576.8
IND-C003	BG	tr	A0A149PFH2	A0A149PFH2_9BURK	RHD	564.6
IND-C004	BG	tr	A0ABR9P133	A0ABR9P133_9ACTN	RHD	562.3
IND-C005	BG	tr	A0AA37H9X9	A0AA37H9X9_9HYPH	RHD	562.3
IND-C006	BG	tr	A0ABT1QJY3	A0ABT1QJY3_9NOCA	RHD	558.4
IND-C007	BG	tr	A0A238K6P7	A0A238K6P7_9RHOB	RHD	558.0
IND-C008	BG	tr	A0A076NQM4	A0A076NQM4_9CORY	RHD	557.1
IND-C009	BG	tr	A0A3N6MW60	A0A3N6MW60_9BURK	RHD	556.4
IND-C010	BG	tr	A0A4P8HLQ0	A0A4P8HLQ0_9BURK	RHD	556.2
IND-C011	BG	tr	A0A1M4V8D6	A0A1M4V8D6_9GAMM	RHD	554.7
IND-C012	BG	tr	A0ABQ2DPT0	A0ABQ2DPT0_9MICC	RHD	554.2
IND-C013	BG	tr	A0A101UVV8	A0A101UVV8_9ACTN	RHD	554.0
IND-C014	BG	tr	A0A059ZN19	A0A059ZN19_ACIBA	RHD	553.9
IND-C015	BG	tr	A0ABP8PRI8	A0ABP8PRI8_9NOCA	RHD	553.9
IND-C016	BG	tr	A0A1H2MWH9	A0A1H2MWH9_9PSED	RHD	553.8
IND-C017	BG	tr	A0A438BKP0	A0A438BKP0_9NOCA	RHD	552.0
IND-C018	BG	tr	Q2T826	Q2T826_BURTA	RHD	551.9

IND-C019	BG	tr	A0ABM7PXD5	A0ABM7PXD5_SINCY	RHD	549.3
IND-C020	BG	tr	A0A939GY85	A0A939GY85_9BURK	RHD	548.3
IND-C021	BG	tr	Q9I0W4	Q9I0W4_PSEAE	RHD	547.9
IND-C022	BG	tr	A0AAW5HU82	A0AAW5HU82_9CORY	RHD	546.0
IND-C023	BG	tr	A0A1V2DTW8	A0A1V2DTW8_9GAMM	RHD	538.5
IND-C024	BG	tr	A0A2N7UJ67	A0A2N7UJ67_9GAMM	RHD	538.3
IND-C025	BG	tr	A0ABY1FKU2	A0ABY1FKU2_9GAMM	RHD	537.4
IND-C026	BG	tr	A0A3P8JKU4	A0A3P8JKU4_KLETE	RHD	536.4
IND-C027	BG	tr	A0A2N0H3Q6	A0A2N0H3Q6_9SPHN	RHD	535.3
IND-C028	BG	tr	A0A846TNL6	A0A846TNL6_9MICC	RHD	530.7
IND-C029	BG	tr	A0A9Q9SNG1	A0A9Q9SNG1_9BURK	RHD	530.5
IND-C030	BG	tr	A0ABW2HF70	A0ABW2HF70_9MICO	RHD	525.3
IND-C031	BG	tr	A0A5D0UJA8	A0A5D0UJA8_9ACTN	RHD	519.9
IND-C032	BG	tr	Q934B6	Q934B6_RHOJR	RHD	518.1
IND-C033	BG	tr	A0A2K2G0K4	A0A2K2G0K4_9SPHN	RHD	505.1
IND-C034	BG	tr	A0A562KKN7	A0A562KKN7_SPHWJ	RHD	492.4
IND-C035	BG	tr	A0ABW2S4T9	A0ABW2S4T9_9NOCA	RHD	490.3
IND-C036	BG	tr	A0ABW3VT00	A0ABW3VT00_9PSEU	RHD	486.0
IND-C037	BG	tr	A0A0B1ZUU5	A0A0B1ZUU5_9SPHN	RHD	477.8
IND-C038	BG	tr	A0ABY1QWW5	A0ABY1QWW5_9SPHN	RHD	477.3
IND-C039	BG	tr	A0ABW0G974	A0ABW0G974_9PROT	RHD	471.5
IND-C040	BG	tr	I0CL65	I0CL65_9BACT	RHD	467.9
IND-C041	BG	tr	A0A852VY74	A0A852VY74_PSEA5	RHD	467.4
IND-C042	BG	tr	A0A1H6J885	A0A1H6J885_MYCRU	RHD	466.7
IND-C043	BG	tr	A0ABX8UE68	A0ABX8UE68_9ACTN	RHD	460.9
IND-C044	BG	tr	A0ABV2WR00	A0ABV2WR00_9NOCA	RHD	457.9
IND-C045	BG	tr	A0A934UTS5	A0A934UTS5_9MICO	RHD	457.8
IND-C046	BG	tr	A0ABQ0KRX9	A0ABQ0KRX9_MYCNV	RHD	457.3
IND-C047	BG	tr	A0ABQ1BL04	A0ABQ1BL04_9MYCO	RHD	448.2
IND-C048	BG	tr	A0ABN6ISY1	A0ABN6ISY1_9MYCO	RHD	446.0
IND-C049	BG	tr	A0A1I5PX20	A0A1I5PX20_9GAMM	RHD	445.5
IND-C050	BG	tr	A0ABP7MVV7	A0ABP7MVV7_9GAMM	HLD	443.0
IND-C051	BG	tr	A0A064CBC8	A0A064CBC8_9MYCO	RHD	428.7
IND-C052	BG	tr	A0A2M9YN64	A0A2M9YN64_9LEPT	HLD	422.5
IND-C053	BG	tr	A0A2U1K4Q3	A0A2U1K4Q3_9BACI	RHD	421.2
IND-C054	BG	tr	A0A7K0CKW3	A0A7K0CKW3_9ACTN	RHD	413.7
IND-C055	BG	tr	A0A543IBY3	A0A543IBY3_9ACTN	HLD	407.3
IND-C056	MGYG000490812	MGYG000490812_01835	HLD	404.8	0.161	
IND-C057	BG	tr	A0A1X1ZSA7	A0A1X1ZSA7_9MYCO	HLD	401.8
IND-C058	MGYG000490814	MGYG000490814_02176	HLD	395.5	0.184	
IND-C059	BG	tr	A0A418MWQ9	A0A418MWQ9_9ACTN	RHD	391.9
IND-C060	BG	tr	A0ABV8I7J0	A0ABV8I7J0_9ACTN	HLD	382.2

Appendix D. Mining run summaries (verbatim JSON)

v1 (contaminated canon)

```
{
  "corpus_total": 61927,
  "mgnify_proteins": 49252,
  "bg_proteins": 1947,
  "decoy_proteins": 8594,
  "candidates": 742,
  "candidates_by_class": {
```

```

    "RHD": 729,
    "HLD": 13
  },
  "candidates_by_source": {
    "bg_rieske_oxy.fasta": 716,
    "bg_dehalogenase_like.fasta": 8,
    "mgnify": 18
  },
  "novel_jacc_lt_0.5": 740,
  "validation": {
    "seed": 20260923,
    "positive_control": {
      "recovered": 82,
      "total": 82
    },
    "gate1_recall_verified_heldout": {
      "recovered": 27,
      "total": 31,
      "recall": 0.871,
      "misses": [
        "P37334",
        "P77650",
        "Q46373",
        "Q8U671"
      ]
    },
    "gate1_all_heldout": {
      "recovered": 31,
      "total": 35
    },
    "gate2": {
      "precision": 1.0,
      "auROC": 0.987,
      "eval_negatives_called": [],
      "n_eval_negatives": 36
    },
    "thresholds_T": {
      "FAMX:RHD:RHD_OTHER": 80.6,
      "FAMX:HLD:HALOALKANE": 230.2,
      "FAMX:RHD:RHD_BPH": 53.3,
      "FAMX:RHD:RHD_BTX": 12.5,
      "FAMX:RHD:RHD_PAH": 834.3,
      "FAMX:HLD:HALOACID": 276.2
    },
    "thresholds_D": {
      "FAMX:RHD:RHD_OTHER": 0.0,
      "FAMX:HLD:HALOALKANE": 0.0,
      "FAMX:RHD:RHD_BPH": 0.0,
      "FAMX:RHD:RHD_BTX": 0.0,
      "FAMX:RHD:RHD_PAH": 0.0,
      "FAMX:HLD:HALOACID": 0.0
    }
  }
}

```

v2 (curated canon)

```

{
  "corpus_total": 61927,
  "mgnify_proteins": 49252,
  "bg_proteins": 1947,
  "decoy_proteins": 8594,

```

```

"candidates": 351,
"candidates_by_class": {
  "RHD": 338,
  "HLD": 13
},
"candidates_by_source": {
  "bg_rieske_oxy.fasta": 336,
  "bg_dehalogenase_like.fasta": 8,
  "mgnify": 7
},
"novel_jacc_lt_0.5": 349,
"validation": {
  "seed": 20260923,
  "positive_control": {
    "recovered": 61,
    "total": 61
  },
  "gate1_recall_verified_heldout": {
    "recovered": 18,
    "total": 20,
    "recall": 0.9,
    "misses": [
      "Q8G8B6",
      "Q8U671"
    ]
  },
  "gate1_all_heldout": {
    "recovered": 26,
    "total": 29
  },
  "gate2": {
    "precision": 1.0,
    "auROC": 1.0,
    "eval_negatives_called": [],
    "n_eval_negatives": 36
  },
  "thresholds_T": {
    "FAMX:RHD:RHD_OTHER": 136.0,
    "FAMX:HLD:HALOALKANE": 230.2,
    "FAMX:RHD:RHD_BPH": 555.7,
    "FAMX:RHD:RHD_BTX": 631.6,
    "FAMX:RHD:RHD_PAH": 842.1,
    "FAMX:HLD:HALOACID": 276.2
  },
  "thresholds_D": {
    "FAMX:RHD:RHD_OTHER": 0.0,
    "FAMX:HLD:HALOALKANE": 0.0,
    "FAMX:RHD:RHD_BPH": 0.0,
    "FAMX:RHD:RHD_BTX": 0.0,
    "FAMX:RHD:RHD_PAH": 0.0,
    "FAMX:HLD:HALOACID": 0.0
  }
}
}

```

Appendix E. MGnify MAG provenance (16 MAGs)

MGnify accession	Catalogue	Proteins
MGYG000490834	soil-v1-0	3350
MGYG000490818	soil-v1-0	3114
MGYG000490830	soil-v1-0	4571

MGYG000490820	soil-v1-0	3108
MGYG000490809	soil-v1-0	4089
MGYG000490812	soil-v1-0	3408
MGYG000490804	soil-v1-0	3601
MGYG000490814	soil-v1-0	4367
MGYG000518840	marine-sediment-v1-0	3091
MGYG000518823	marine-sediment-v1-0	3372
MGYG000518835	marine-sediment-v1-0	1588
MGYG000518825	marine-sediment-v1-0	1923
MGYG000518809	marine-sediment-v1-0	2927
MGYG000518815	marine-sediment-v1-0	1301
MGYG000518804	marine-sediment-v1-0	3844
MGYG000518820	marine-sediment-v1-0	1598

References

1. EnzymeMiner 2.0. Nucleic Acids Research 2025.
<https://academic.oup.com/nar/article/54/W1/W257/8675561>
2. Metagenome-derived haloalkane dehalogenases with novel catalytic properties. 2017.
<https://pubmed.ncbi.nlm.nih.gov/28674849/>
3. Characterization of novel PAH dioxygenases from bacterial metagenomic DNA of contaminated soil. AEM 2014. <https://doi.org/10.1128/aem.01883-14>
4. UniProtKB REST API. <https://rest.uniprot.org/uniprotkb/>
5. MGnify genome catalogues soil-v1-0, marine-sediment-v1-0.
<https://www.ebi.ac.uk/metagenomics/>
6. pyhmmer (HMMER3 Python bindings). <https://pypi.org/project/pyhmmer/>
7. pesticides/ slice, this repo: GATES_LOCKED.md, paper_draft.pdf (platform template).
8. Janssen DB et al., haloalkane dehalogenase mechanism (DhlA, Xanthobacter). Canon accession P22643, <https://www.uniprot.org/uniprotkb/P22643/entry>